Mathematical Induction Problems

1. Prove that for $n! > 2^n$; for all $n \geq 4$.
2. Prove that for any integer $n \geq 1$, $2^{2n} - 1$ is divisible by 3.
3. Let a and b be two distinct integers, prove that $a^n - b^n$ are divisible by $a - b$.
4. Prove that for all integers n, $\left(1+\frac{1}{1^3}\right)\left(1+\frac{1}{2^3}\right) \dots \dots \left(1+\frac{1}{n^3}\right) < 3$.
5. Prove that F_{3n} is even for every $n \geq 1$.
6. Prove that F_{4n} is divisible by three for every $n \geq 1$.
7. Show that $1 < \frac{F_n}{F_{n-1}} < 2$ for all $n > 3$.
8. Prove that for every $n, k \in N$: $F_n \leq F_k F_{n-k} + F_{k+1} F_{n-k-1} \leq F_{n+1}$
9. Prove that every positive integer n, there exists an n- digit number divisible by 5^n all of whose digits are odd.
10. Show that, $ab^n + cn + d$ is divisible by the positive integer m, given that $a + d, (b-1)c$ & $ab - a + c$ are divisible by m.
11. Prove that every natural number can be represented as a sum of several distinct powers of 2.
12. Prove that for any natural number n, $2^{3^n} + 1$ is divisible by 3^{n+1}.
13. Let α be any real number such that $\alpha + \frac{1}{\alpha} \in Z$. Prove that $\alpha^n + \frac{1}{\alpha^n} \in Z$, for any $n \in N$.
14. For all $n \in N$, we have $f(n) = g(n)$, where:

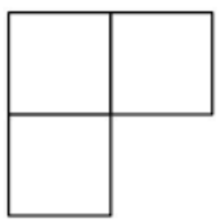
$$f(n) = 1 - \frac{1}{2} + \frac{1}{3} - \cdots + \frac{1}{2n-1} - \frac{1}{2n}$$

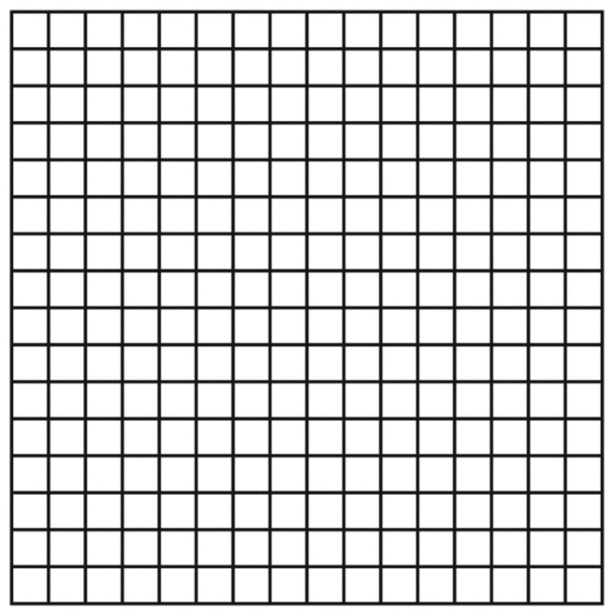
$$g(n) = \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n}$$

15. For any natural number N, prove the inequality

$$\sqrt{2\sqrt{3\sqrt{4\ldots\ldots\sqrt{N-1\sqrt{N}}}}} < 3$$

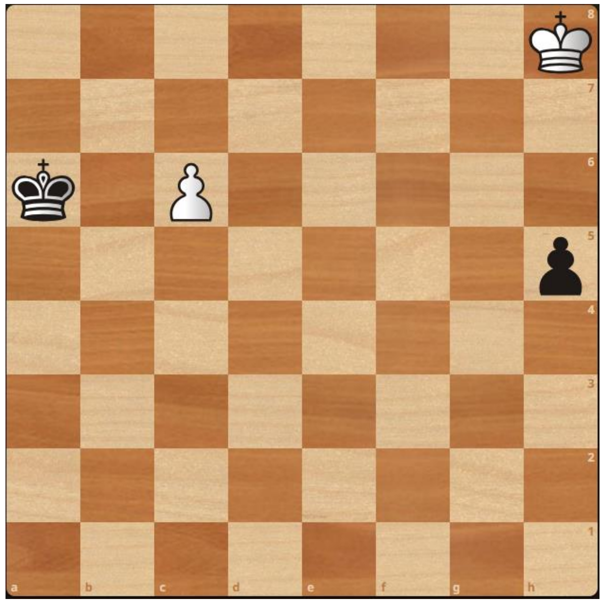
16. One box was cut off from a 16x16 square of graph paper. Prove that the figure obtained can be dissected into trominos of a certain type- “corners”. (See blow figure)



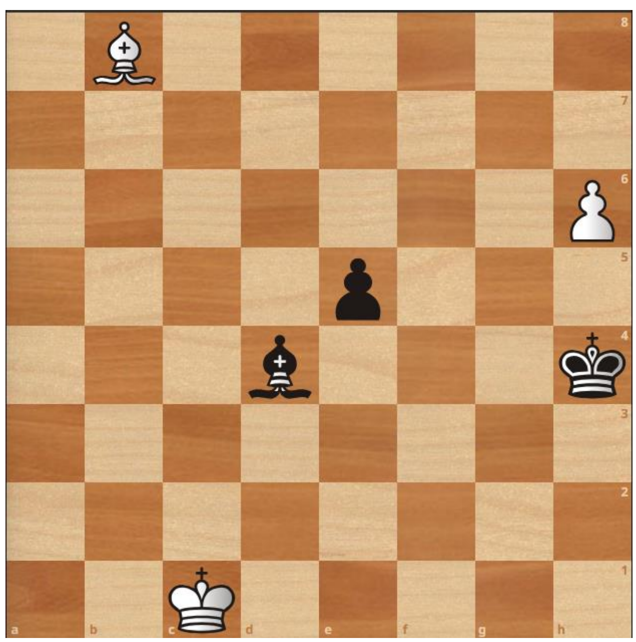


Chess Problems

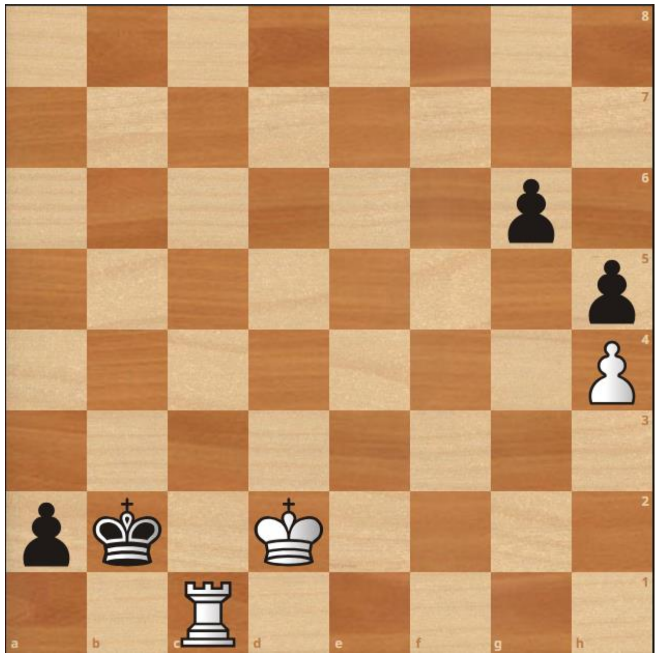
1. White to Play and Draw



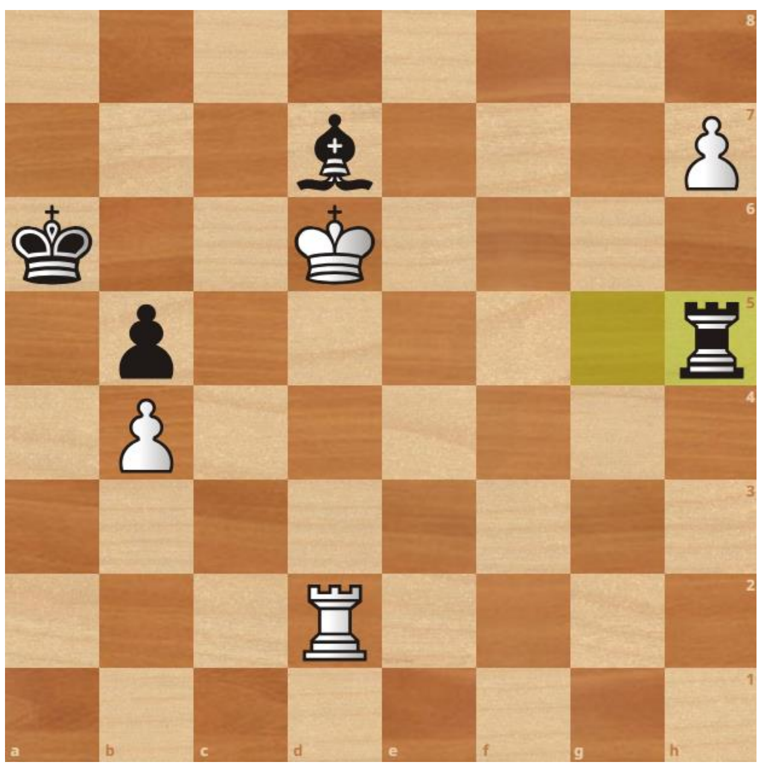
2. White to Play and Win



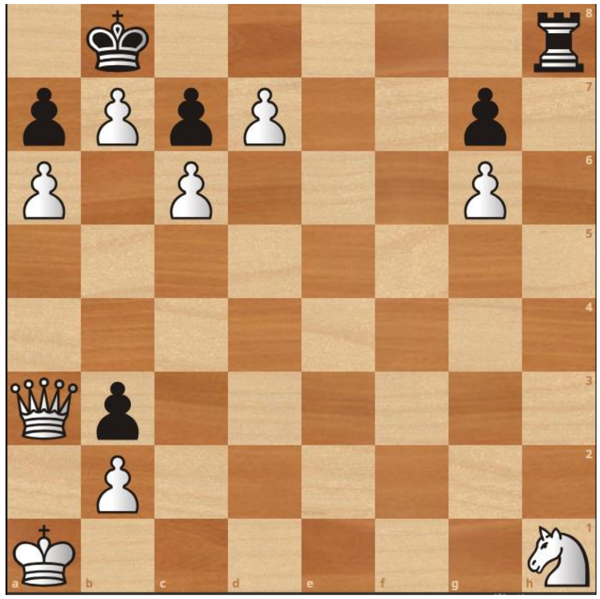
3. White to Play and Win



4. White to Play and Win



5. White to Play and Win



6. White to Play and Win